

Beam pruning Ranking Preservation under Bounded Score Perturbations

Research Architecture Runtime
operator/beam-pruning

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Abstract

`beam-pruning` is certified by pairwise ranking preservation when all gaps exceed 2ε . Lean `LEAN_FULL` via `OrderStat.Ranking`.

This theorem is: Reduction.

1 Problem

The `beam-pruning` operator depends on the pairwise order of scores $s \in \mathbb{R}^n$. The certified object is the full pairwise ranking (and thereby any ranking-determined selection such as top- k index sets).

2 Stability notion

Perturbations: $\|\delta\|_\infty \leq \varepsilon$. Stability: if `AllGapsExceed`($s, 2\varepsilon$), then $s_i < s_j \Leftrightarrow (s + \delta)_i < (s + \delta)_j$ for all i, j . Sharpness: a gap $\leq 2\varepsilon$ admits a collision adversary.

3 Definitions

Definition 1 (All-gaps). $|s_p - s_q| > g$ for $p \neq q$.

Assumptions. $n \geq 2$, $\varepsilon \geq 0$, and (for invariance) all pairwise gaps $> 2\varepsilon$.

4 Theorem

Theorem 2 (Ranking invariance). *If `AllGapsExceed`($s, 2\varepsilon$), then for every $\|\delta\|_\infty \leq \varepsilon$ and all i, j , $s_i < s_j \Leftrightarrow s_i + \delta_i < s_j + \delta_j$.*

Theorem 3 (Sharpness). *If some distinct pair has $|s_i - s_j| \leq 2\varepsilon$, an admissible δ forces a value collision.*

5 Intuition

Relative ℓ_∞ moves cost at most 2ε . Gaps larger than 2ε cannot reverse; smaller gaps can be closed by a midpoint push.

6 Examples

Example 4. Scores $(0, 3, 8)$, $\varepsilon = 1$: all gaps > 2 , so the ranking is stable. If scores are $(0, 1.5)$ and $\varepsilon = 1$, the gap $1.5 \leq 2$ admits a midpoint tie.

7 Proof

Fix $\|\delta\|_\infty \leq \varepsilon$. If $i = j$ both sides of the claimed equivalence are false for strict $<$. If $i \neq j$ and $|s_i - s_j| > 2\varepsilon$, the estimate $(s_i + \delta_i) - (s_j + \delta_j) \geq (s_i - s_j) - 2\varepsilon$ (and its swap) yields $s_i < s_j \Leftrightarrow s_i + \delta_i < s_j + \delta_j$ (Theorem 2).

For sharpness, take $i \neq j$ with $|s_i - s_j| \leq 2\varepsilon$ and the midpoint $\delta_i = -(s_i - s_j)/2$, $\delta_j = (s_i - s_j)/2$ (elsewhere 0). Then $\|\delta\|_\infty \leq \varepsilon$ and $s_i + \delta_i = s_j + \delta_j$ (Theorem 3).

8 Formal statement

Lean module: `Research.Operators.BeamPruning.Preservation`. Source file: `lean/Research/Operators/BeamPruning`.
Theorems: `beam_pruning_margin_invariance`, `beam_pruning_margin_sharpness`. Reduction of
`Research.Operators.OrderStat.Ranking`. Certificate: `lean/certificates/beam-pruning/beam-pruning-margin`.
Status: LEAN_FULL on Mathlib $\mathbb{R}$ (REAL_MATHLIB).

9 Proof dependencies

- `Research.Operators.OrderStat.Ranking` / shared pairwise lemma from `KthMargin`.
- Midpoint collision adversary.
- No other operator theorems required.

10 Consequences

Ranking stability implies stability of any functional of the pairwise order used by `beam-pruning`.
Related operators: `top-k`, `sorting`, `rank`.